Vertical Jump Coach Agent — Progress Report

WatSPEED Agentic AI · Course Project · Sean Tang · due Nov 8, 2026
Code: github.com/seantanger/thp-jump-coach-agent (branch `main`)

For this project, I am building an AI coach for vertical jump training based on the information from John Evans / THP Strength YouTube video transcripts. The work I have done is with the help of Claude Code. The agents use the Claude API. The first thing I did was collect all of the transcripts from YouTube using yt-dlp, and preprocess the text with regex rules. I then tried three chunking techniques: fixed-size character splitting (LangChain's `RecursiveCharacterTextSplitter`) at two chunk sizes, and a third method that cuts the transcript into ~60-second windows at natural pauses in speech. After embedding the chunks using nomic embed in Ollama, I made a simple LangGraph loop where the Coach calls a search tool itself (up to 4 searches, rewriting its query if needed) and answers with the video and timestamp. I compared each of the three chunking methods with three search methods (keyword/BM25, vector, and a hybrid of the two) on 30 test questions. The best combination was the larger character chunks with vector search. I also built a second agent, a Planner, that writes multi-week training programs from reviewed templates, runs them through code checks, and waits for my approval before saving (this is still a WIP).

I went through a few versions of the Coach prompt, most recently to make answers more concise. For safeguards, a code check rejects any answer that cites a source the search didn't return, a third agent (the Verifier) checks each point against its source in a fresh context, and a code safety check sends messages with red-flag pain symptoms (like a pop in the knee) to a physio instead of coaching them.

Next Steps/Ideas:

- Continue working on the planning agent
- Connect the Planner and Coach. Right now they are two separate LangGraph graphs: a code router detects plan requests, but the chat doesn't pass them to the Planner yet. I want to maybe combine them into one parent graph with the Coach and Planner as subgraphs.
- Give the Coach access to the current plan and profile, so its answers fit where the athlete is in the program.
- Memory: right now the profile and each plan version are saved as JSON files. Would something like Letta be better?
- Training log: record sessions as done, changed or missed, plus pain and jump results, and use them to decide when the athlete moves to the next stage. (Not sure

if this should be like a UI or the user should enter a chat to tell the system that they have finished a workout)
- Workout variety: the Planner can now swap exercises (at most 2 per week, each cited and equipment-checked); next is more options per slot and loosening sets and reps.
- Optional: MCP server.

Demos

These call the Claude API (about 4¢ per Coach question) and read the transcript index. The outputs are saved in this notebook.

In [1]:
```
import os, sys
from pathlib import Path

ROOT = (Path.cwd().parent if Path.cwd().name == "notebooks" else Path.cwd())
os.chdir(ROOT); sys.path.insert(0, str(ROOT))
os.environ.setdefault("RETRIEVAL_MODE", "dense")  # dense (vector) search ne
```

Out[1]:
```
'dense'
```

The Coach's LangGraph setup (`src/coach/graph.py`). Two nodes: `coach` (the LLM with tools bound) and `tools` (runs the searches, capped at 4). The graph loops coach → tools → coach until the LLM stops calling tools. `answer_question` (used below) builds this graph, runs it, checks the citations, and sends the answer to the Verifier.

1. Ask the Coach a question. The Coach searches the library, answers with citations, then the Verifier checks it. (The two agents are linked through code, not the LangGraph)

In [2]:
```
from src.coach.qa import answer_question

r = answer_question("Should I static stretch before jumping?")
print(r["compact"])
print("\nsearches:", r["queries"])
print("citation check passed:", r["check"]["passed"], "| Verifier:", r["chec
```

```
Generally no. Skip static stretching right before jumping and use dynamic fl
exibility instead. If you need it for mobility, do short holds and follow th
em with something dynamic or ballistic, or do them after the session.

- Static stretching before jumping can temporarily lower force and power out
put, so keep the pre-jump warm-up dynamic (RDLs, side squats, leg swings, lo
ng lunges). [1][2]
- The power drop is dose-dependent. Long, intense stretching hurts most, and
30–60 seconds followed by something ballistic largely cancels the negative e
ffect. [3][4]
- If tight ankles or hips limit your squat or jump positions, do static stre
tches after your warm-up, or in the evening. [5][4]

**Sources**
1. [John Evans — A Practical 2 Day Per Week Jump System Guaranteed To Get Yo
u Dunking @ 01:42](https://www.youtube.com/watch?v=1sQ-FOxDP5M&t=102s)
2. [THP Strength — World Record Vertical Jumper's Exact Warm-Up Routine (50.
5 Inches) @ 00:51](https://www.youtube.com/watch?v=O4piVLyfvSA&t=51s)
3. [John Evans — Does Stretching ACTUALLY Make You A Better Athlete? What Do
es Science Say? @ 00:00](https://www.youtube.com/watch?v=ifrQ8lcxoNs&t=0s)
4. [THP Strength — This Test Will Show You If You Can Reach A 40 Inch Vertic
al @ 07:44](https://www.youtube.com/watch?v=8l0Omkm28hE&t=464s)
5. [John Evans — A Practical 2 Day Per Week Jump System Guaranteed To Get Yo
u Dunking @ 05:42](https://www.youtube.com/watch?v=1sQ-FOxDP5M&t=342s)

searches: ['static stretching in the warm-up before jumping', 'static stretc
hing effect on jump performance power output']
citation check passed: True | Verifier: pass
```

A pain question: the answer sticks to what the library says about tendon pain, says what it doesn't cover, and ends with the physio line. Note that regular AI models typically recommend ice and anti inflammatories which are NOT recommended by THP.

In [3]:
```
r = answer_question("What should I do if I have knee pain?")
print(r["compact"])
print("\nsearches:", r["queries"])
print("citation check passed:", r["check"]["passed"], "| Verifier:", r["chec
```

```
If your knee pain is activity-related tendon ache under the kneecap, take 2
4–48 hours off, then track your morning pain. If it stays above 3/10, start
daily isometrics.

- For an early or reactive flare-up, taking 24 to 48 hours off often settles
the tendon. You can then return to normal activity without new exercises.
[1]
- Test pain each morning with a deep bodyweight squat, scored 1–10. Pain pea
ks 24–48 hours after a jump session. If the score is above 3/10, start isome
trics. [2][3]
- Isometrics: hold a quad exercise such as a leg extension for 45 seconds at
about 70% effort. Do 3–5 sets, about three times a day, spaced roughly 6 hou
rs apart. [4][5]

*See a physio if pain is sharp/stabbing or steadily worsening, occurs at res
t or wakes you at night, lasts over two weeks without improving despite cutt
ing load, or follows a fall/impact; or if you have swelling, bruising, defor
mity, a pop, giving way, locking, catching, numbness, tingling, radiating we
akness or limping. Back pain with bladder/bowel changes or groin numbness ne
eds emergency care.*

**Sources**
1. [THP Strength – How to Fix Patellar Tendon Pain Based on Your Tendinopath
y Phase @ 01:24](https://www.youtube.com/watch?v=iLI8hXsB6hs&t=84s)
2. [THP Strength – How to Fix Patellar Tendon Pain Without Quitting Your Spo
rt @ 03:07](https://www.youtube.com/watch?v=ZN8wh4FFZuA&t=187s)
3. [THP Strength – The Exact Protocol to Fix Knee Pain and Get Back to Jumpi
ng @ 00:00](https://www.youtube.com/watch?v=VMSF8C25mxY&t=0s)
4. [John Evans – How To Get Rid Of Knee Pain (Full Blueprint) @ 05:28](http
s://www.youtube.com/watch?v=9X_j-oF2ZDg&t=328s)
5. [THP Strength – Knee Pain Killing Your Vertical? Start With Isometrics @
02:17](https://www.youtube.com/watch?v=TMznIxcRJGg&t=137s)

searches: ["patellar tendon pain tendinopathy jumper's knee what to do", 'kn
ee pain training load modification rehab isometrics']
citation check passed: True | Verifier: pass
```

2. Safety check (code, no LLM). Red-flag symptoms get a physio referral before any agent runs; plan requests are flagged for the Planner (not yet wired into the chat).

In [4]:
```
from src.red_flags import referral_kind
from src.router import route
```

In [5]:
```
for m in ["Should I static stretch before jumping?", "I felt a pop in my kne
          "Make me a plan for the next 12 weeks"]:
    print(f"{m:42} referral={referral_kind(m)}  route={route(m)}")
```

```
Should I static stretch before jumping?    referral=None  route=coach
I felt a pop in my knee this week          referral=physio  route=coach
Make me a plan for the next 12 weeks       referral=None  route=program
```